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	Departamento Department	Stress Methods (MESAT)	Avión Aircraft
<u>Titulo/Title</u> Program ALLOWEE, version 4 User's manual			
<u>Palabras clave/Key words</u> ALLOWABLES, COMPOSITE MATERIALS, SOFTWARE			Clasificación acceso Access class P1 Registro de Revisiones Revision Record Pag. 2
<u>Resumen/Summary</u> This technical report describes how to use the ALLOWEE Computer Program, Version 4, and it includes examples to further clarify its use. The ALLOWEE Program calculates the allowable values (A Basis value and B Basis value) of composite material properties as it is shown in the MIL-HDBK-17-1D procedure. This program is part of section XVIII (Computer Programs and Procedures) of Volume 8 (Stress Analysis) of C.A.S.A.'s Design Manual.			
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1 Technical sheet of the Program

Nota Técnica **ME-T-NT-130109**

PROGRAM

ALLOWEE

REFERENCE

DATE

Febrero 2003

OBJECTIVE

The ALLOWEE program calculates the allowable values (A Basis value and B Basis value) of composite material properties as it is shown in the MIL-HDBK-17-1D procedure.

LANGUAGE

Fortran 77

Visual Basic .NET

COMPUTER

Sun/PC

PC

NUMBER OF LINES (APROX.)

3900 (Fortran, analysis core)

+3000 (Visual Basic .Net)

EXECUTION MODE

Interactive

Interactive GUI

Total 6900

Notes:

- Blank lines and commented lines excluded
- Count for Visual Basic code excludes lines created by Visual Studio in designer.vb files

AUTHOR

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VERSION

1.0

S. Cervigón

2.0

L. Claramonte

2.1

A. Matesanz

3.0

J.A.Velasco

4.0

ADDITIONAL DOCUMENTS

Nota Técnica NT-T-ADP-08003:

Manual Teórico del Programa ALLOWEE, versión 3.

Nota Técnica NT-T-ADP-03005:

Manual del Programador del Programa ALLOWEE.

2 Introduction

2.1 Object

This technical report describes how to use the ALLOWEE Computer Program, version 4, and it includes examples to further clarify its use.

The ALLOWEE Program calculates the allowable values (A Basis value and B Basis value) of composite material properties following the method explained in Ref. 1. Moreover, the program does the following calculations:

- For more than one batch:
 - It determines if the data are from the same population (k-sample Anderson-Darling test).
 - It determines if the different batches have similar variances (Levene test).
- It determines if the data fit in any of these distributions: Weibull, normal or lognormal.
- It calculates the parameters and A and B basis value of the distributions Weibull, normal and lognormal, and of the methods ANOVA and non parametric.
- It finds 'outliers' in the different batches and, if the values are poolable, in the whole population. But all the calculations are done using all the data.
- If the data are from different environmental conditions, the program determines if it is possible to pool the data and, in that case, it calculates the allowable values for each environmental condition.

2.2 References

Ref. 1 NT-T-ADP-08003 Ed. A "Manual Teórico del Programa ALLOWEE, versión 3"

Ref. 2 NT-T-ADP-03005 Ed. A "Manual del Programador del Programa ALLOWEE"

Ref. 3 PP0300181 Is. 2 "Composite statistical methods to determine B-bases value from small data sets".

2.3 Differences among versions

Version 4.0

The user can introduce the Coefficient of Variation of the samples population, which will be set by the program as a lower bound of the CV value to be used in the Small Samples Formula (ignored in the rest of the statistical methods).

3 Program Execution

3.1 PC Server Version

An application has been developed in Visual Basic for Windows environment. The original routines of the program for calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore, the same results will be obtained with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed, for installation, please contact with the system manager.

The application can be executed by "double click" in the Allowee.exe, located in the PC's server, or by means a **link** in the "desktop" or in the "start menu"

<\\Cae\InfoCalculo\Soft\Allowee\>

In the folder <\\Cae\InfoCalculo\Soft\Allowee\Consola> , there is another executable program, which is the Fortran program in console mode.

In the PC version, when the program starts, the following information window is displayed:



Figure 3.1. Program start

The input data must be typed in the application form or retrieved from an input file. The Form, after loading the data corresponding to the example 1 (see section §7), has the following appearance:

Allowee

CASA DENOMINATION **PROBLEM 1 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **M**

VARIATION COEFFICIENT (%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

No. of batches: **9** ☐ There are different environmental conditions

Type of data	No. of data	1	2	3	4	5	6	7
1 XXXXXX	7	65.4	63.6	63.4	61.3	61	61.6	65.4
2 XXXXXX	5	64	64.8	66.4	61.2	64.7		
3 XXXXXX	5	68.4	68.8	66.8	63.6	69.3		
4 XXXXXX	7	66.8	67.9	68.9	65.6	64.6	71.1	64.1
5 XXXXXX	6	74.6	74.2	62.7	73.3	68.6	74	
6 XXXXXX	5	67.4	68	74.6	74.9	65.5		
7 XXXXXX	5	69.4	71.5	69.9	86.4	65.4		
8 XXXXXX	6	64.7	70	70.2	79.2	76.8	78.5	
9 XXXXXX	7	70.7	80.6	68.3	87	81.1	79.7	82.3

Language **Input file** **Output file** **Execute** **Return to data** **Exit**

Figure 3.2. Input data form

The program can be executed clicking the button "Program execution". Then the application solves the problems showing the following form:

Allowee

CASA DENOMINATION **PROBLEM 1 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **M**

VARIATION COEFFICIENT (%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

Allowables **Statistics of each batch** **Results Weibull and Normal** **Results Lognormal and ANOVA** **Outliers**

A-Basis Value **45.7**

B-Basis Value **55.7**

Remarks

- ANOVA method.
- Batches with different distributions.
- Batches with similar variances.
- There are outliers.

Language **Input file** **Output file** **Execute** **Return to data** **Exit**

Figure 3.3. Output results form

The output results shown in this form can also be written in the output file (details in section §5).

3.2 Multivac Computer Version

To execute the ALLOWEE program in the MULTIVAC computer, its full path name has to be typed unless the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table), it can be executed by typing the command:

\$allowee

Data input is performed by means of a data file. The dialog with the program is the following:

<u>Program ALLOWEE</u> This program calculates the allowable values (A and B) of the mechanical properties of composites materials using the procedure of MIL-HDBK-17-1D.	
Version 4.0	June 2013

To continue, press <RETURN>

Notes

- This software and the data that are produced by it are EADS-CASA property.
The total or partial reproduction of this software is forbidden, as well as its use out of the EADS-CASA environment.

To continue, press <RETURN>

The input data file must be introduced interactively via the keyboard:

Name of the input data file	→
-----------------------------	---

The full name of the file containing the output data must be given.

Name of the output file without extension [ALLOWEE] →

4 Input data file

The composition of the input data file is the following:

```

DEN CASA: CASA_den
DEN MAT.: Mat_den
TEST:      Test_den
UNITS:      Unit
LOTS:       N_b
CV(%):      CVP
'Data_type'(1), N_d(1), Data1(1), Data2(1), Data3(1), ..., DataNd(1)}(1)
-----
'Data_type'(i), N_d(i), Data1(i), Data2(i), Data3(i), ..., DataNd(i)}(i)
-----
-----
'Data_type'(N_b), N_d(N_b), Data1(N_b), Data2(N_b), Data3(N_b), ..., DataNd(N_b)}(N_b)
Calculation data type
Environmental conditions
    
```

The symbols used are:

CASA_den

Denomination CASA of the material used.

Mat_den

Denomination of the material used.

Test_den

Denomination of the type and conditions of the test done.

Unit

Units in which the data are given.

N_b

Number of batches whose data will be given next. It must be an integer.

CVP

Coefficient of Variation of the samples population, in %. This magnitude is used only in the Small Samples Formula: the calculation of the B-value is done with the maximum between this figure (CVP) and the CV calculated for the sample data. The recommended values of CVP for Fibre Reinforced Polymer materials are (see Ref. 3):

- Young Modulus 4.7 %
- Fiber strength (in-plane tension) 5.8 %
- Resin strength (compression, shear, CAI,...) 9.0 %

The reading of the previous data is done from the first character different from a blank after the ':'. They must be given in one line and they must have a maximum of 255 characters.

Next, the values of the data obtained of each of the N_b batches are given. One line per batch is written with the following values:

'Data_type' (i)

It identifies the data type of the batch i . If the batch is from the laboratory reception then a letter 'L' must be written inside the character 'Data_type', any position is possible. If the batch is from qualification then a letter 'C' must be written inside the character 'Data_type', any position is possible. If it doesn't matter where the values are from because all the data will be used in the calculations, it is not necessary to write anything here.

N_d(i)

Number of data of the batch i .

Data₁(i), Data₂(i), ..., Data_{N_d(i)}(i)

Values obtained for the batch i .

Calculation data type

It is an integer that identifies the data used in the calculation. If a number '1' is written here only the batches from laboratory reception must be used in the calculations. If a number '2' is written here only the batches from qualification must be used in the calculations. If a number '3' or anything except '1' or '2' is written here all values are used in the calculations.

Environmental conditions

It identifies if data are from different environmental conditions or not. If '**Y**' or '**y**' (yes) is written here, the program will consider that each batch is from different environmental condition. If '**N**' or '**n**' (no) or anything except '**Y**' or '**y**' is written here different environmental conditions are not consider.

If all the values are going to be used in calculations and there aren't different environmental conditions, it is not necessary to write anything after the data of the batches. But if the user wants to consider only some values or there are data from different environmental conditions, both lines must be written.

5 Output results file

The output '.RES' file is structured in several groups:

1. Analysis identification

- CASA denomination
- Material denomination
- Test name and conditions
- Units
- Total number of data values
- Batches number
- Coefficient of Variation of the samples population

2. Allowable values and remarks

- A and B allowable values; if the data are from different environmental conditions and they can be poolable, here the A and B basis value for each environmental condition is given.
- Remarks
 - If there are batches from different environmental conditions, it is said if these data are poolable or not.
 - Distribution assumed for the data or method used.
 - If the data are from the same population or not.
 - If the batches have similar variances or not.
 - If outliers exist.

3. Input data

- Batch number
- Number of values of each batch
- Individual values of each batch

4. Statistics of each batch

- Mean values of each batch
- Standard deviation of each batch

- Mean of the whole sample
- Standard deviation of the whole sample

4. *Batch to batch variability* (only if the batch number is bigger than one)

- Results of the k-simple Anderson-Darling test
- Results of the equality of variances test

5. *Summary of allowables* (only in big samples)

- Two parameters Weibull method: parameters, tolerance limit factors, observance level and A and B values if the population is considered to fit a Weibull distribution.
- Normal method: parameters, tolerance limit factors, observance level and A and B values if the population is considered to fit a Normal distribution.
- Lognormal method: parameters, tolerance limit factors, observance level and A and B values if the population is considered to fit a Lognormal distribution.
- Non parametrical method: A and B values if the non parametrical method is used.
- ANOVA method: parameters, tolerance limit factors, observance level and A and B values if ANOVA method is used.

6. *Small sample additional data* (only in small samples)

- Kb of the normalized data: calculated using the normalized data and assuming normal distribution.
- Coefficient of variation used: it only appears when the small sample formula is used and, in that case, it is the maximum value between the coefficient of variation calculated for the sample data, and the coefficient of variation of the samples population (defined in the input file).
- Kb obtained from the small sample formula: it only appears when the small sample formula is used and it is calculated as it was shown in the theoretical manual.

6 Limitations of the program

There are some intrinsic limitations in the calculation method used in this program. Those limitations are notified if they occur by warnings on the screen or in the output file. They are the following:

- The non-parametrical method can be only used if there isn't an important variation between batches.
- The ANOVA method cannot be applied if there are less than 3 batches.
- Allowable values cannot be calculated by the two parameters Weibull method when there are less than 10 values in the sample.
- Allowables cannot be calculated for less than 4 values.
- The Levene test of equality of variances cannot be applied if the variance of any of the samples is null.

Moreover those limitations in the calculation methods, there are some more requirements in the Military Handbook MIL-HDBK-17-1D about the sample size in order that the allowables values may be published like B-basis values.

- The sample size necessary to publish the calculated allowables values as B-basis values depends on the material condition and its specification in the process. If there isn't a normalized specification of the production process, it is necessary a minimum of 5 batches with 6 specimens each from each manufacturer that wants to publish a B-basis value. If there is a normalized specification of the production process, it is possible to pool the data from each manufacturer that wants to publish a B-basis value. In that case, it is required a minimum of 3 manufacturers with 3 batches each manufacturer and with 6 specimens each batch.
- The specimens of the samples must be obtained from random selected panel areas. These panels must be manufactured as it is said in the material specification. The specimen number of each batch must be as uniform as it will be possible, and the size of the biggest batch must not be more than one and a half times bigger than the size of the smallest batch.

If the previous requirements are not fulfilled, the B-basis value can be calculated but the results cannot be published.

7 Examples

Below are 11 examples of the use of this program. The examples one to seven are from the Military Handbook MIL-HDBK-17-1D. It is hoped that these examples will clarify the information given in this manual.

EXAMPLE 1:

The input data file (***ej_allowee_1.dat***) is the following:

DEN CASA: PROBLEM 1 MIL-HDBK-17-1D Section 8.7

DEN MAT.: Z-19776+Z19785

TEST: LAP SHEAR/DIN 65148(RT)

UNITS: MPA

LOTS: 9

CV (%): 2

'XXXXXX', 7, 65.4, 63.6, 63.4, 61.3, 61.0, 61.6, 65.4

'XXXXXX', 5, 64.0, 64.8, 66.4, 61.2, 64.7

'XXXXXX', 5, 68.4, 68.8, 66.8, 63.6, 69.3

'XXXXXX', 7, 66.8, 67.9, 68.9, 65.6, 64.6, 71.1, 64.1

'XXXXXX', 6, 74.6, 74.2, 62.7, 73.3, 68.6, 74.0

'XXXXXX', 5, 67.4, 68.0, 74.6, 74.9, 65.5

'XXXXXX', 5, 69.4, 71.5, 69.9, 86.4, 65.4

'XXXXXX', 6, 64.7, 70.0, 70.2, 79.2, 76.8, 78.5

'XXXXXX', 7, 70.7, 80.6, 68.3, 87.0, 81.1, 79.7, 82.3

The output file is:

ej_allowee_1.RES

18-06-2013, Allowee 4.0

CASA DENOMINATION : PROBLEM 1 MIL-HDBK-17-1D Section 8.7
 MATERIAL DENOMINATION : Z-19776+Z19785
 TEST : LAP SHEAR/DIN 65148(RT)
 UNITS : MPA
 TOTAL NUMBER OF VALUES : 53
 NUMBER OF BATCHES : 9

ALLOWABLE VALUES

A-BASIS VALUE = 45.74
 B-BASIS VALUE = 55.67

REMARKS:

ANOVA method.
 Batches with different distributions. The data should not be pooled.
 Batches with similar variances.
 There are outliers. Outliers are included in all calculations.

INPUT DATA

BATCH	N	SPECIMEN DATA					
1	7	61.0	61.3	61.6	63.4	63.6	65.4
		65.4					
2	5	61.2	64.0	64.7	64.8	66.4	
3	5	63.6	66.8	68.4	68.8	69.3	
4	7	64.1	64.6	65.6	66.8	67.9	68.9
		71.1					
5	6	62.7	68.6	73.3	74.0	74.2	74.6
6	5	65.5	67.4	68.0	74.6	74.9	
7	5	65.4	69.4	69.9	71.5	86.4 *	
8	6	64.7	70.0	70.2	76.8	78.5	79.2
9	7	68.3	70.7	79.7	80.6	81.1	82.3
		87.0					

TOTAL OF 9 BATCHES, 53 SPECIMEN(S).

* = Outlier at the 0.05 level.

=====

PRELIMINARY DESCRIPTIVE STATISTICS

NUMBER OF BATCH	SAMPLE SIZE	AVERAGE	STANDARD DEVIATION
1	7	63.10	1.86
2	5	64.22	1.90
3	5	67.38	2.31
4	7	67.00	2.50
5	6	71.23	4.73
6	5	70.08	4.36
7	5	72.52	8.08
8	6	73.23	5.81
9	7	78.53	6.64

POOLED SAMPLE	53	69.78	6.48

=====

INVESTIGATION OF BATCH-TO-BATCH VARIABILITY

THE K-SAMPLE ANDERSON-DARLING TEST STATISTIC = 2.67
 THE CRITICAL VALUE = 1.44
 SIGNIFICANT LEVEL = 0.05
 DIFFERENT POPULATION

THE TEST STATISTIC FOR EQUALITY OF VARIANCE = 1.05
 THE CRITICAL VALUE WITH 9-1 DEGREES OF FREEDOM = 2.16
 SIGNIFICANT LEVEL = 0.05
 VARIANCES ARE NOT SIGNIFICANTLY DIFFERENT

=====

COMPUTATION ALLOWABLE VALUES

THE TWO-PARAMETER WEIBULL METHOD

SHAPE PARAMETER = 10.21
 SCALE PARAMETER = 72.86
 TOLERANCE LIMIT FACTOR (A Basis) = 8.38
 TOLERANCE LIMIT FACTOR (B Basis) = 4.65
 A-D STATISTIC = 2.31
 OBSERVANCE LEVEL = 0.00

THE A VALUE = 52.22
 THE B VALUE = 54.91

THE NORMAL METHOD

MEAN = 69.78
 STANDARD DEVIATION = 6.48
 TOLERANCE LIMIT FACTOR (A Basis) = 2.84
 TOLERANCE LIMIT FACTOR (B Basis) = 1.63

A-D STATISTIC	=	1.37
OBSERVANCE LEVEL	=	0.00

THE A VALUE	=	51.35
THE B VALUE	=	59.19

THE LOGNORMAL METHOD

AVERAGE OF LOG DATA	=	4.24
STD. DEV. OF LOG DATA	=	0.09
TOLERANCE LIMIT FACTOR (A Basis)	=	2.84
TOLERANCE LIMIT FACTOR (B Basis)	=	1.63
A-D STATISTIC	=	1.03
OBSERVANCE LEVEL	=	0.01

THE A VALUE	=	53.83
THE B VALUE	=	60.01

THE NONPARAMETRIC METHOD

THE A VALUE	=	49.63	(Hanson-Koopmans value)
THE B VALUE	=	61.20	

NORMAL ANALYSIS IN THE PRESENCE OF BATCH-TO-BATCH VARIATION (ANOVA)

THE ASSUMPTIONS ARE:

- 1) THE DATA FROM EACH BATCH ARE NORMALLY DISTRIBUTED,
- 2) THE BATCH MEANS ARE NORMALLY DISTRIBUTED, AND
- 3) THE VARIANCES WITHIN EACH BATCH ARE THE SAME.

BETWEEN BATCH MEAN SQUARE	=	150.97
WITHIN BATCH VARIANCE	=	22.18
POPULATION STANDARD DEVIATION	=	6.64
TOLERANCE LIMIT FACTOR (A Basis)	=	3.62
TOLERANCE LIMIT FACTOR (B Basis)	=	2.12

THE A VALUE	=	45.74
THE B VALUE	=	55.67

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

B1 Allowee

CASA DENOMINATION **PROBLEM 1 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

No. of batches: **9** ☐ There are different environmental conditions

Type of data	No. of data	1	2	3	4	5	6	7
1 XXXXXX	7	65.4	63.6	63.4	61.3	61	61.6	65.4
2 XXXXXX	5	64	64.8	66.4	61.2	64.7		
3 XXXXXX	5	68.4	68.8	66.8	63.6	69.3		
4 XXXXXX	7	66.8	67.9	68.9	65.6	64.6	71.1	64.1
5 XXXXXX	6	74.6	74.2	62.7	73.3	68.6	74	
6 XXXXXX	5	67.4	68	74.6	74.9	65.5		
7 XXXXXX	5	69.4	71.5	69.9	86.4	65.4		
8 XXXXXX	6	64.7	70	70.2	79.2	76.8	78.5	
9 XXXXXX	7	70.7	80.6	68.3	87	81.1	79.7	82.3

Language:

Input file:

Output file:

Execute Return to data Exit

Figure 7.1. Example 1 input data

If the 'Calculate' button is pressed the results are shown:

B1 Allowee

CASA DENOMINATION **PROBLEM 1 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

Allowables Statistics of each batch Results Weibull and Normal Results Lognormal and ANOVA Outliers

A-Basis Value 45.7

B-Basis Value 55.7

Remarks

- ANOVA method.
- Batches with different distributions.
- Batches with similar variances.
- There are outliers.

Language:

Input file:

Output file:

Execute Return to data Exit

Figure 7.2. Example 1 output results

EXAMPLE 2:

The input data file (***ej_allowee_2.dat***) is the following:

DEN CASA: PROBLEM 2 MIL-HDBK-17-1D Section 8.7

DEN MAT.: Z-19776+Z19785

TEST: LAP SHEAR/DIN 65148(RT)

UNITS: MPA

LOTS: 9

CV. (%): 2

'XXXXXX', 7, 61.3, 68.5, 62.5, 66.0, 66.6, 64.8, 69.5

'XXXXXX', 5, 66.5, 64.7, 64.9, 65.2, 70.3

'XXXXXX', 5, 66.0, 72.7, 67.1, 67.7, 65.7

'XXXXXX', 7, 61.9, 68.0, 63.3, 74.6, 66.2, 68.2, 69.1

'XXXXXX', 6, 68.9, 65.0, 70.9, 65.4, 66.5, 64.9

'XXXXXX', 5, 75.8, 75.2, 71.5, 69.6, 66.1

'XXXXXX', 5, 72.8, 75.0, 66.3, 69.5, 71.9

'XXXXXX', 6, 71.9, 71.0, 69.5, 69.5, 72.6, 74.6

'XXXXXX', 7, 68.7, 76.3, 76.6, 69.6, 66.2, 72.4, 72.8

The output file is:

ej_allowee_2.RES

18-06-2013, Allowee 4.0

CASA DENOMINATION : PROBLEM 2 MIL-HDBK-17-1D Section 8.7
 MATERIAL DENOMINATION : Z-19776+Z19785
 TEST : LAP SHEAR/DIN 65148(RT)
 UNITS : MPA
 TOTAL NUMBER OF VALUES : 53
 NUMBER OF BATCHES : 9

ALLOWABLE VALUES

A-BASIS VALUE = 55.50
 B-BASIS VALUE = 61.05

REMARKS:

ANOVA method.
 Batches with different distributions. The data should not be pooled.
 Batches with similar variances.
 There are not outliers.

INPUT DATA

BATCH	N	SPECIMEN DATA					
1	7	61.3	62.5	64.8	66.0	66.6	68.5
		69.5					
2	5	64.7	64.9	65.2	66.5	70.3	
3	5	65.7	66.0	67.1	67.7	72.7	
4	7	61.9	63.3	66.2	68.0	68.2	69.1
		74.6					
5	6	64.9	65.0	65.4	66.5	68.9	70.9
6	5	66.1	69.6	71.5	75.2	75.8	
7	5	66.3	69.5	71.9	72.8	75.0	
8	6	69.5	69.5	71.0	71.9	72.6	74.6
9	7	66.2	68.7	69.6	72.4	72.8	76.3
		76.6					

TOTAL OF 9 BATCHES, 53 SPECIMEN(S).

PRELIMINARY DESCRIPTIVE STATISTICS

NUMBER OF BATCH	SAMPLE SIZE	AVERAGE	STANDARD DEVIATION
1	7	65.60	2.99
2	5	66.32	2.33
3	5	67.84	2.84
4	7	67.33	4.17
5	6	66.93	2.45
6	5	71.64	4.03
7	5	71.10	3.33
8	6	71.52	1.96
9	7	71.80	3.88

POOLED SAMPLE	53	68.84	3.87

INVESTIGATION OF BATCH-TO-BATCH VARIABILITY

```

THE K-SAMPLE ANDERSON-DARLING TEST STATISTIC    =    1.81
THE CRITICAL VALUE                               =    1.44
SIGNIFICANT LEVEL                               =    0.05
DIFFERENT POPULATION

THE TEST STATISTIC FOR EQUALITY OF VARIANCE       =    0.57
THE CRITICAL VALUE WITH 9-1 DEGREES OF FREEDOM   =    2.16
SIGNIFICANT LEVEL                               =    0.05
VARIANCES ARE NOT SIGNIFICANTLY DIFFERENT
    
```

COMPUTATION ALLOWABLE VALUES

THE TWO-PARAMETER WEIBULL METHOD

```

SHAPE PARAMETER                                =    18.75
SCALE PARAMETER                                =    70.68
TOLERANCE LIMIT FACTOR (A Basis)               =    8.38
TOLERANCE LIMIT FACTOR (B Basis)               =    4.65
A-D STATISTIC                                  =    1.06
OBSERVANCE LEVEL                              =    0.01

THE A VALUE                                    =    58.96
THE B VALUE                                    =    60.59
    
```

THE NORMAL METHOD

```

MEAN                                            =    68.84
STANDARD DEVIATION                            =    3.87
TOLERANCE LIMIT FACTOR (A Basis)              =    2.84
TOLERANCE LIMIT FACTOR (B Basis)              =    1.63
A-D STATISTIC                                  =    0.51
    
```

OBSERVANCE LEVEL = 0.18

THE A VALUE = 57.84

THE B VALUE = 62.52

THE LOGNORMAL METHOD

AVERAGE OF LOG DATA = 4.23

STD. DEV. OF LOG DATA = 0.06

TOLERANCE LIMIT FACTOR (A Basis) = 2.84

TOLERANCE LIMIT FACTOR (B Basis) = 1.63

A-D STATISTIC = 0.43

OBSERVANCE LEVEL = 0.26

THE A VALUE = 58.62

THE B VALUE = 62.73

THE NONPARAMETRIC METHOD

THE A VALUE = 53.86 (Hanson-Koopmans value)

THE B VALUE = 61.90

NORMAL ANALYSIS IN THE PRESENCE OF BATCH-TO-BATCH VARIATION (ANOVA)

THE ASSUMPTIONS ARE:

- 1) THE DATA FROM EACH BATCH ARE NORMALLY DISTRIBUTED,
- 2) THE BATCH MEANS ARE NORMALLY DISTRIBUTED, AND
- 3) THE VARIANCES WITHIN EACH BATCH ARE THE SAME.

BETWEEN BATCH MEAN SQUARE = 39.64

WITHIN BATCH VARIANCE = 10.47

POPULATION STANDARD DEVIATION = 3.93

TOLERANCE LIMIT FACTOR (A Basis) = 3.39

TOLERANCE LIMIT FACTOR (B Basis) = 1.98

THE A VALUE = 55.50

THE B VALUE = 61.05

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

B1 Allowee

CASA DENOMINATION **PROBLEM 2 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

No. of batches: **9** ☐ There are different environmental conditions

	Type of data	No. of data	1	2	3	4	5	6	7
1	XXXXXX	7	61.3	68.5	62.5	66	66.6	64.8	69.5
2	XXXXXX	5	66.5	64.7	64.9	65.2	70.3		
3	XXXXXX	5	66	72.7	67.1	67.7	65.7		
4	XXXXXX	7	61.9	68	63.3	74.6	66.2	68.2	69.1
5	XXXXXX	6	68.9	65	70.9	65.4	66.5	64.9	
6	XXXXXX	5	75.8	75.2	71.5	69.6	66.1		
7	XXXXXX	5	72.8	75	66.3	69.5	71.9		
8	XXXXXX	6	71.9	71	69.5	69.5	72.6	74.6	
9	XXXXXX	7	68.7	76.3	76.6	69.6	66.2	72.4	72.8

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.3. Example 2 input data

If the 'Calculate' button is pressed the results are shown:

B1 Allowee

CASA DENOMINATION **PROBLEM 2 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

Allowables **Statistics of each batch** Results Weibull and Normal Results Lognormal and ANOVA

Batch no.	Sample	Mean	Standar dev.	CoV (%)
1	7	65.60	2.99	4.55
2	5	66.32	2.33	3.52
3	5	67.84	2.84	4.18
4	7	67.33	4.17	6.19
5	6	66.93	2.45	3.66
6	5	71.64	4.03	5.62
7	5	71.10	3.33	4.68
8	6	71.52	1.96	2.74
9	7	71.80	3.88	5.41

No. of batches **9**

Total no. of data **53**

Mean **68.84**

Standar deviation **3.87**

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.4. Example 2 output results

EXAMPLE 3:

The input data file (**ej_allowee_3.dat**) is the following:

DEN CASA: PROBLEM 3 MIL-HDBK-17-1D Section 8.7

DEN MAT.: Z-19776+Z19785

TEST: LAP SHEAR/DIN 65148(RT)

UNITS: MPA

LOTS: 9

CV (%): 2

'XXXXXX', 7, 67.4, 61.8, 72.6, 56.0, 69.0, 75.0, 65.4

'XXXXXX', 5, 59.4, 66.2, 74.8, 75.7, 63.9

'XXXXXX', 5, 70.5, 61.7, 66.6, 66.1, 72.7

'XXXXXX', 7, 65.5, 61.7, 60.6, 75.0, 64.4, 75.3, 71.7

'XXXXXX', 6, 51.6, 64.3, 74.2, 75.4, 60.5, 47.1

'XXXXXX', 5, 73.5, 77.0, 66.7, 66.4, 65.0

'XXXXXX', 5, 64.0, 66.2, 64.3, 60.4, 72.4

'XXXXXX', 6, 71.0, 66.9, 76.8, 64.4, 70.0, 72.1

'XXXXXX', 7, 73.9, 78.5, 68.0, 75.0, 79.5, 60.6, 71.4

The output file is:

ej_allowee_3.RES

18-06-2013, Allowee 4.0

CASA DENOMINATION : PROBLEM 3 MIL-HDBK-17-1D Section 8.7
 MATERIAL DENOMINATION : Z-19776+Z19785
 TEST : LAP SHEAR/DIN 65148(RT)
 UNITS : MPA
 TOTAL NUMBER OF VALUES : 53
 NUMBER OF BATCHES : 9

ALLOWABLE VALUES

A-BASIS VALUE = 53.63
 B-BASIS VALUE = 55.92

REMARKS:

Data fit in a Weibull distribution.
 Batches are from the same population. The data may be pooled.
 Batches with similar variances.
 There are not outliers.

INPUT DATA

BATCH	N	SPECIMEN DATA					
1	7	56.0	61.8	65.4	67.4	69.0	72.6
		75.0					
2	5	59.4	63.9	66.2	74.8	75.7	
3	5	61.7	66.1	66.6	70.5	72.7	
4	7	60.6	61.7	64.4	65.5	71.7	75.0
		75.3					
5	6	47.1	51.6	60.5	64.3	74.2	75.4
6	5	65.0	66.4	66.7	73.5	77.0	
7	5	60.4	64.0	64.3	66.2	72.4	
8	6	64.4	66.9	70.0	71.0	72.1	76.8
9	7	60.6	68.0	71.4	73.9	75.0	78.5
		79.5					

TOTAL OF 9 BATCHES, 53 SPECIMEN(S).

PRELIMINARY DESCRIPTIVE STATISTICS

NUMBER OF BATCH	SAMPLE SIZE	AVERAGE	STANDARD DEVIATION
1	7	66.74	6.45
2	5	68.00	7.06
3	5	67.52	4.26
4	7	67.74	6.18
5	6	62.18	11.54
6	5	69.72	5.24
7	5	65.46	4.41
8	6	70.20	4.30
9	7	72.41	6.53
POOLED SAMPLE	53	67.85	6.77

INVESTIGATION OF BATCH-TO-BATCH VARIABILITY

THE K-SAMPLE ANDERSON-DARLING TEST STATISTIC = 0.99
 THE CRITICAL VALUE = 1.44
 SIGNIFICANT LEVEL = 0.05
 SAME POPULATION

THE TEST STATISTIC FOR EQUALITY OF VARIANCE = 1.30
 THE CRITICAL VALUE WITH 9-1 DEGREES OF FREEDOM = 2.16
 SIGNIFICANT LEVEL = 0.05
 VARIANCES ARE NOT SIGNIFICANTLY DIFFERENT

COMPUTATION ALLOWABLE VALUES

THE TWO-PARAMETER WEIBULL METHOD

SHAPE PARAMETER = 12.27
 SCALE PARAMETER = 70.77
 TOLERANCE LIMIT FACTOR (A Basis) = 8.38
 TOLERANCE LIMIT FACTOR (B Basis) = 4.65
 A-D STATISTIC = 0.54
 OBSERVANCE LEVEL = 0.17

 THE A VALUE = 53.63
 THE B VALUE = 55.92

THE NORMAL METHOD

MEAN = 67.85
 STANDARD DEVIATION = 6.77
 TOLERANCE LIMIT FACTOR (A Basis) = 2.84

TOLERANCE LIMIT FACTOR (B Basis) = 1.63
 A-D STATISTIC = 0.56
 OBSERVANCE LEVEL = 0.13

THE A VALUE = 48.62
 THE B VALUE = 56.80

THE LOGNORMAL METHOD

AVERAGE OF LOG DATA = 4.21
 STD. DEV. OF LOG DATA = 0.10
 TOLERANCE LIMIT FACTOR (A Basis) = 2.84
 TOLERANCE LIMIT FACTOR (B Basis) = 1.63
 A-D STATISTIC = 0.76
 OBSERVANCE LEVEL = 0.04

THE A VALUE = 50.16
 THE B VALUE = 56.92

THE NONPARAMETRIC METHOD

THE A VALUE = 34.75 (Hanson-Koopmans value)
 THE B VALUE = 51.60

NORMAL ANALYSIS IN THE PRESENCE OF BATCH-TO-BATCH VARIATION (ANOVA)

THE ASSUMPTIONS ARE:

- 1) THE DATA FROM EACH BATCH ARE NORMALLY DISTRIBUTED,
- 2) THE BATCH MEANS ARE NORMALLY DISTRIBUTED, AND
- 3) THE VARIANCES WITHIN EACH BATCH ARE THE SAME.

BETWEEN BATCH MEAN SQUARE = 53.37
 WITHIN BATCH VARIANCE = 44.38
 POPULATION STANDARD DEVIATION = 6.78
 TOLERANCE LIMIT FACTOR (A Basis) = 2.91
 TOLERANCE LIMIT FACTOR (B Basis) = 1.68

THE A VALUE = 48.10
 THE B VALUE = 56.48

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

B1 Allowee

CASA DENOMINATION **PROBLEM 3 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

No. of batches: **9** ☐ There are different environmental conditions

	Type of data	No. of data	1	2	3	4	5	6	7
1	XXXXXX	7	67.4	61.8	72.6	56	69	75	65.4
2	XXXXXX	5	59.4	66.2	74.8	75.7	63.9		
3	XXXXXX	5	70.5	61.7	66.6	66.1	72.7		
4	XXXXXX	7	65.5	61.7	60.6	75	64.4	75.3	71.7
5	XXXXXX	6	51.6	64.3	74.2	75.4	60.5	47.1	
6	XXXXXX	5	73.5	77	66.7	66.4	65		
7	XXXXXX	5	64	66.2	64.3	60.4	72.4		
8	XXXXXX	6	71	66.9	76.8	64.4	70	72.1	
9	XXXXXX	7	73.9	78.5	68	75	79.5	60.6	71.4

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.5. Example 3 input data

If the 'Calculate' button is pressed the results are shown:

B1 Allowee

CASA DENOMINATION **PROBLEM 3 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

Allowables **Statistics of each batch** **Results Weibull and Normal** **Results Lognormal and ANOVA**

A-Basis Value 53.6

B-Basis Value 55.9

Remarks

- Data fit in a Weibull distribution.
- Batches are from the same population.
- Batches with similar variances.
- There are not outliers.

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.6. Example 3 output results

EXAMPLE 4:

The input data file (**ej_allowee_4.dat**) is the following:

DEN CASA: PROBLEM 4 MIL-HDBK-17-1D Section 8.7

DEN MAT.: Z-19776+Z19785

TEST: LAP SHEAR/DIN 65148(RT)

UNITS: MPA

LOTS: 9

CV (%): 2

'XXXXXX', 7, 30.4, 29.6, 29.3, 27.9, 29.8, 30.5, 29.4

'XXXXXX', 5, 31.5, 31.1, 27.9, 34.5, 29.5

'XXXXXX', 5, 29.6, 29.8, 29.1, 30.0, 30.4

'XXXXXX', 7, 32.1, 29.7, 26.9, 30.0, 31.4, 28.3, 31.9

'XXXXXX', 6, 30.0, 31.5, 32.0, 30.2, 30.6, 31.1

'XXXXXX', 5, 29.6, 29.3, 30.4, 32.1, 32.8

'XXXXXX', 5, 29.9, 30.9, 32.1, 31.4, 32.0

'XXXXXX', 6, 29.2, 31.0, 31.8, 30.3, 28.2, 28.0

'XXXXXX', 7, 30.1, 29.2, 27.7, 28.0, 31.4, 29.6, 32.3

The output file is:

ej_allowee_4.RES

18-06-2013, Allowee 4.0

CASA DENOMINATION : PROBLEM 4 MIL-HDBK-17-1D Section 8.7
 MATERIAL DENOMINATION : Z-19776+Z19785
 TEST : LAP SHEAR/DIN 65148(RT)
 UNITS : MPA
 TOTAL NUMBER OF VALUES : 53
 NUMBER OF BATCHES : 9

ALLOWABLE VALUES

A-BASIS VALUE = 26.00
 B-BASIS VALUE = 27.81

=====

REMARKS:

Data fit in a Normal distribution.
 Batches are from the same population. The data may be pooled.
 Batches with similar variances.
 There are not outliers.

=====

INPUT DATA

BATCH	N	SPECIMEN DATA					

1	7	27.9	29.3	29.4	29.6	29.8	30.4
		30.5					
2	5	27.9	29.5	31.1	31.5	34.5	
3	5	29.1	29.6	29.8	30.0	30.4	
4	7	26.9	28.3	29.7	30.0	31.4	31.9
		32.1					
5	6	30.0	30.2	30.6	31.1	31.5	32.0
6	5	29.3	29.6	30.4	32.1	32.8	
7	5	29.9	30.9	31.4	32.0	32.1	
8	6	28.0	28.2	29.2	30.3	31.0	31.8
9	7	27.7	28.0	29.2	29.6	30.1	31.4
		32.3					

TOTAL OF 9 BATCHES, 53 SPECIMEN(S).

PRELIMINARY DESCRIPTIVE STATISTICS

NUMBER OF BATCH	SAMPLE SIZE	AVERAGE	STANDARD DEVIATION
1	7	29.56	0.87
2	5	30.90	2.47
3	5	29.78	0.48
4	7	30.04	1.94
5	6	30.90	0.77
6	5	30.84	1.54
7	5	31.26	0.90
8	6	29.75	1.54
9	7	29.76	1.68

POOLED SAMPLE	53	30.25	1.50

INVESTIGATION OF BATCH-TO-BATCH VARIABILITY

THE K-SAMPLE ANDERSON-DARLING TEST STATISTIC = 1.04
 THE CRITICAL VALUE = 1.44
 SIGNIFICANT LEVEL = 0.05
 SAME POPULATION

THE TEST STATISTIC FOR EQUALITY OF VARIANCE = 1.77
 THE CRITICAL VALUE WITH 9-1 DEGREES OF FREEDOM = 2.16
 SIGNIFICANT LEVEL = 0.05
 VARIANCES ARE NOT SIGNIFICANTLY DIFFERENT

COMPUTATION ALLOWABLE VALUES

THE TWO-PARAMETER WEIBULL METHOD

SHAPE PARAMETER = 20.33
 SCALE PARAMETER = 30.96
 TOLERANCE LIMIT FACTOR (A Basis) = 8.38
 TOLERANCE LIMIT FACTOR (B Basis) = 4.65
 A-D STATISTIC = 0.86
 OBSERVANCE LEVEL = 0.02

 THE A VALUE = 26.19
 THE B VALUE = 26.86

THE NORMAL METHOD

MEAN = 30.25
 STANDARD DEVIATION = 1.50
 TOLERANCE LIMIT FACTOR (A Basis) = 2.84
 TOLERANCE LIMIT FACTOR (B Basis) = 1.63

A-D STATISTIC	=	0.36
OBSERVANCE LEVEL	=	0.36

THE A VALUE	=	26.00
THE B VALUE	=	27.81

THE LOGNORMAL METHOD

AVERAGE OF LOG DATA	=	3.41
STD. DEV. OF LOG DATA	=	0.05
TOLERANCE LIMIT FACTOR (A Basis)	=	2.84
TOLERANCE LIMIT FACTOR (B Basis)	=	1.63
A-D STATISTIC	=	0.38
OBSERVANCE LEVEL	=	0.34

THE A VALUE	=	26.25
THE B VALUE	=	27.87

THE NONPARAMETRIC METHOD

THE A VALUE	=	23.28	(Hanson-Koopmans value)
THE B VALUE	=	27.70	

NORMAL ANALYSIS IN THE PRESENCE OF BATCH-TO-BATCH VARIATION (ANOVA)

THE ASSUMPTIONS ARE:

- 1) THE DATA FROM EACH BATCH ARE NORMALLY DISTRIBUTED,
- 2) THE BATCH MEANS ARE NORMALLY DISTRIBUTED, AND
- 3) THE VARIANCES WITHIN EACH BATCH ARE THE SAME.

BETWEEN BATCH MEAN SQUARE	=	2.43
WITHIN BATCH VARIANCE	=	2.20
POPULATION STANDARD DEVIATION	=	1.50
TOLERANCE LIMIT FACTOR (A Basis)	=	2.88
TOLERANCE LIMIT FACTOR (B Basis)	=	1.66

THE A VALUE	=	25.94
THE B VALUE	=	27.77

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

Allowee

CASA DENOMINATION **PROBLEM 4 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

No. of batches: **9** ☐ There are different environmental conditions

Type of data	No. of data	1	2	3	4	5	6	7
1	7	30.4	29.6	29.3	27.9	29.8	30.5	29.4
2	5	31.5	31.1	27.9	34.5	29.5		
3	5	29.6	29.8	29.1	30	30.4		
4	7	32.1	29.7	26.9	30	31.4	28.3	31.9
5	6	30	31.5	32	30.2	30.6	31.1	
6	5	29.6	29.3	30.4	32.1	32.8		
7	5	29.9	30.9	32.1	31.4	32		
8	6	29.2	31	31.8	30.3	28.2	28	
9	7	30.1	29.2	27.7	28	31.4	29.6	32.3

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.7. Example 4 input data

If the 'Calculate' button is pressed the results are shown:

Allowee

CASA DENOMINATION **PROBLEM 4 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

Allowables **Statistics of each batch** Results Weibull and Normal Results Lognormal and ANOVA

LogNormal

Mean of log data **3.41**

Standard deviation of log data **0.05**

Tolerance Limit factor (A Basis) **2.84**

Tolerance Limit factor (B Basis) **1.63**

A-D statistics **0.38**

Observation level **0.34**

A Value **26.25**

B Value **27.87**

ANOVA

Between batch mean square **2.4**

Within batch variance **2.2**

Population standard deviation **1.5**

Tolerance Limit factor (A Basis) **2.88**

Tolerance Limit factor (B Basis) **1.66**

A Value **25.94**

B Value **27.77**

Nonparametric

A Value **23.28**

B Value **27.70**

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.8. Example 4 output results

EXAMPLE 5:

The input data file (**ej_allowee_5.dat**) is the following:

DEN CASA: PROBLEM 5 MIL-HDBK-17-1D Section 8.7

DEN MAT.: Z-19776+Z19785

TEST: LAP SHEAR/DIN 65148(RT)

UNITS: MPA

LOTS: 9

CV (%): 2

'XXXXXX', 7, 30.4, 29.0, 34.8, 30.5, 27.3, 27.9, 29.4

'XXXXXX', 5, 29.8, 32.4, 33.2, 29.9, 30.5

'XXXXXX', 5, 31.1, 32.0, 31.0, 29.3, 29.9

'XXXXXX', 7, 32.2, 29.9, 29.1, 29.9, 32.7, 31.6, 31.6

'XXXXXX', 6, 27.5, 29.3, 29.1, 29.5, 28.7, 29.5

'XXXXXX', 5, 29.9, 28.9, 29.2, 29.1, 31.1

'XXXXXX', 5, 31.1, 30.4, 33.4, 30.0, 27.3

'XXXXXX', 6, 31.3, 30.2, 30.2, 29.1, 33.0, 34.4

'XXXXXX', 7, 28.4, 30.1, 30.5, 29.4, 28.7, 32.3, 29.6

The output file is:

ej_allowee_5.RES

18-06-2013, Allowee 4.0

CASA DENOMINATION : PROBLEM 5 MIL-HDBK-17-1D Section 8.7
 MATERIAL DENOMINATION : Z-19776+Z19785
 TEST : LAP SHEAR/DIN 65148(RT)
 UNITS : MPA
 TOTAL NUMBER OF VALUES : 53
 NUMBER OF BATCHES : 9

ALLOWABLE VALUES

A-BASIS VALUE = 25.90
 B-BASIS VALUE = 27.68

REMARKS:

Data fit in a Lognormal distribution.
 Batches are from the same population. The data may be pooled.
 Batches with similar variances.
 There are not outliers.

INPUT DATA

BATCH	N	SPECIMEN DATA					
1	7	27.3	27.9	29.0	29.4	30.4	30.5
		34.8					
2	5	29.8	29.9	30.5	32.4	33.2	
3	5	29.3	29.9	31.0	31.1	32.0	
4	7	29.1	29.9	29.9	31.6	31.6	32.2
		32.7					
5	6	27.5	28.7	29.1	29.3	29.5	29.5
6	5	28.9	29.1	29.2	29.9	31.1	
7	5	27.3	30.0	30.4	31.1	33.4	
8	6	29.1	30.2	30.2	31.3	33.0	34.4
9	7	28.4	28.7	29.4	29.6	30.1	30.5
		32.3					

TOTAL OF 9 BATCHES, 53 SPECIMEN(S).

PRELIMINARY DESCRIPTIVE STATISTICS

NUMBER OF BATCH	SAMPLE SIZE	AVERAGE	STANDARD DEVIATION
1	7	29.90	2.46
2	5	31.16	1.55
3	5	30.66	1.06
4	7	31.00	1.36
5	6	28.93	0.76
6	5	29.64	0.90
7	5	30.44	2.19
8	6	31.37	1.99
9	7	29.86	1.30

POOLED SAMPLE	53	30.31	1.68
=====			

INVESTIGATION OF BATCH-TO-BATCH VARIABILITY

THE K-SAMPLE ANDERSON-DARLING TEST STATISTIC = 1.24
 THE CRITICAL VALUE = 1.44
 SIGNIFICANT LEVEL = 0.05
 SAME POPULATION

THE TEST STATISTIC FOR EQUALITY OF VARIANCE = 0.76
 THE CRITICAL VALUE WITH 9-1 DEGREES OF FREEDOM = 2.16
 SIGNIFICANT LEVEL = 0.05
 VARIANCES ARE NOT SIGNIFICANTLY DIFFERENT

COMPUTATION ALLOWABLE VALUES

THE TWO-PARAMETER WEIBULL METHOD

SHAPE PARAMETER = 17.14
 SCALE PARAMETER = 31.14
 TOLERANCE LIMIT FACTOR (A Basis) = 8.38
 TOLERANCE LIMIT FACTOR (B Basis) = 4.65
 A-D STATISTIC = 1.91
 OBSERVANCE LEVEL = 0.00

 THE A VALUE = 25.53
 THE B VALUE = 26.31

THE NORMAL METHOD

MEAN = 30.31
 STANDARD DEVIATION = 1.68
 TOLERANCE LIMIT FACTOR (A Basis) = 2.84
 TOLERANCE LIMIT FACTOR (B Basis) = 1.63
 A-D STATISTIC = 0.81

OBSERVANCE LEVEL = 0.03

THE A VALUE = 25.52

THE B VALUE = 27.56

THE LOGNORMAL METHOD

AVERAGE OF LOG DATA = 3.41

STD. DEV. OF LOG DATA = 0.05

TOLERANCE LIMIT FACTOR (A Basis) = 2.84

TOLERANCE LIMIT FACTOR (B Basis) = 1.63

A-D STATISTIC = 0.65

OBSERVANCE LEVEL = 0.08

THE A VALUE = 25.90

THE B VALUE = 27.68

THE NONPARAMETRIC METHOD

THE A VALUE = 23.71 (Hanson-Koopmans value)

THE B VALUE = 27.30

NORMAL ANALYSIS IN THE PRESENCE OF BATCH-TO-BATCH VARIATION (ANOVA)

THE ASSUMPTIONS ARE:

- 1) THE DATA FROM EACH BATCH ARE NORMALLY DISTRIBUTED,
- 2) THE BATCH MEANS ARE NORMALLY DISTRIBUTED, AND
- 3) THE VARIANCES WITHIN EACH BATCH ARE THE SAME.

BETWEEN BATCH MEAN SQUARE = 3.82

WITHIN BATCH VARIANCE = 2.66

POPULATION STANDARD DEVIATION = 1.69

TOLERANCE LIMIT FACTOR (A Basis) = 2.99

TOLERANCE LIMIT FACTOR (B Basis) = 1.72

THE A VALUE = 25.27

THE B VALUE = 27.40

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

B1 Allowep

CASA DENOMINATION **PROBLEM 5 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

No. of batches: **9** ☐ There are different environmental conditions

Type of data	No. of data	1	2	3	4	5	6	7
1	7	30.4	29	34.8	30.5	27.3	27.9	29.4
2	5	29.8	32.4	33.2	29.9	30.5		
3	5	31.1	32	31	29.3	29.9		
4	7	32.2	29.9	29.1	29.9	32.7	31.6	31.6
5	6	27.5	29.3	29.1	29.5	28.7	29.5	
6	5	29.9	28.9	29.2	29.1	31.1		
7	5	31.1	30.4	33.4	30	27.3		
8	6	31.3	30.2	30.2	29.1	33	34.4	
9	7	28.4	30.1	30.5	29.4	28.7	32.3	29.6

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.9. Example 5 input data

If the 'Calculate' button is pressed the results are shown:

B1 Allowep

CASA DENOMINATION **PROBLEM 5 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

Allowables | Statistics of each batch | **Results Weibull and Normal** | Results Lognormal and ANOVA

Weibull		Normal	
Shape parameter	17.14	Mean	30.31
Scale parameter	31.14	Standard deviation	1.68
Tolerance Limit factor (A Basis)	8.38	Tolerance Limit factor (A Basis)	2.84
Tolerance Limit factor (B Basis)	4.65	Tolerance Limit factor (B Basis)	1.63
A-D statistics	1.91	A-D statistics	0.81
Observation level	0.00	Observation level	0.03
A Value	25.53	A Value	25.52
B Value	26.31	B Value	27.56

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.10. Example 5 output results

EXAMPLE 6:

The input data file (***ej_allowee_6.dat***) is the following:

DEN CASA: PROBLEM 6 MIL-HDBK-17-1D Section 8.7

DEN MAT.: Z-19776+Z19785

TEST: LAP SHEAR/DIN 65148(RT)

UNITS: MPA

LOTS: 9

CV (%): 2

'XXXXXX', 7, 72.3, 65.3, 74.3, 68.5, 65.7, 64.3, 72.8

'XXXXXX', 5, 74.5, 66.3, 74.3, 69.3, 64.9

'XXXXXX', 5, 75.3, 68.5, 75.2, 74.5, 69.6

'XXXXXX', 7, 65.6, 74.6, 76.5, 72.0, 69.2, 74.3, 78.7

'XXXXXX', 6, 68.1, 72.1, 66.6, 75.0, 67.2, 73.4

'XXXXXX', 5, 66.4, 72.6, 67.5, 76.3, 65.1

'XXXXXX', 5, 73.3, 72.6, 67.8, 69.4, 67.3

'XXXXXX', 6, 75.2, 72.8, 67.8, 76.2, 72.8, 68.7

'XXXXXX', 7, 72.3, 76.0, 65.8, 69.9, 67.8, 74.6, 71.3

The output file is:

ej_allowee_6.RES

18-06-2013, Allowee 4.0

CASA DENOMINATION : PROBLEM 6 MIL-HDBK-17-1D Section 8.7
 MATERIAL DENOMINATION : Z-19776+Z19785
 TEST : LAP SHEAR/DIN 65148(RT)
 UNITS : MPA
 TOTAL NUMBER OF VALUES : 53
 NUMBER OF BATCHES : 9

ALLOWABLE VALUES

A-BASIS VALUE = 57.18
 B-BASIS VALUE = 64.90

REMARKS:

Non parametric method.
 Batches are from the same population. The data may be pooled.
 Batches with similar variances.
 There are not outliers.

INPUT DATA

BATCH	N	SPECIMEN DATA					
1	7	64.3	65.3	65.7	68.5	72.3	72.8
		74.3					
2	5	64.9	66.3	69.3	74.3	74.5	
3	5	68.5	69.6	74.5	75.2	75.3	
4	7	65.6	69.2	72.0	74.3	74.6	76.5
		78.7					
5	6	66.6	67.2	68.1	72.1	73.4	75.0
6	5	65.1	66.4	67.5	72.6	76.3	
7	5	67.3	67.8	69.4	72.6	73.3	
8	6	67.8	68.7	72.8	72.8	75.2	76.2
9	7	65.8	67.8	69.9	71.3	72.3	74.6
		76.0					

TOTAL OF 9 BATCHES, 53 SPECIMEN(S).

PRELIMINARY DESCRIPTIVE STATISTICS

NUMBER OF BATCH	SAMPLE SIZE	AVERAGE	STANDARD DEVIATION
1	7	69.03	4.09
2	5	69.86	4.44
3	5	72.62	3.30
4	7	72.99	4.46
5	6	70.40	3.55
6	5	69.58	4.71
7	5	70.08	2.74
8	6	72.25	3.39
9	7	71.10	3.61

POOLED SAMPLE	53	70.91	3.82
=====			

INVESTIGATION OF BATCH-TO-BATCH VARIABILITY

THE K-SAMPLE ANDERSON-DARLING TEST STATISTIC = 0.89
 THE CRITICAL VALUE = 1.44
 SIGNIFICANT LEVEL = 0.05
 SAME POPULATION

THE TEST STATISTIC FOR EQUALITY OF VARIANCE = 0.26
 THE CRITICAL VALUE WITH 9-1 DEGREES OF FREEDOM = 2.16
 SIGNIFICANT LEVEL = 0.05
 VARIANCES ARE NOT SIGNIFICANTLY DIFFERENT

COMPUTATION ALLOWABLE VALUES

THE TWO-PARAMETER WEIBULL METHOD

SHAPE PARAMETER = 21.12
 SCALE PARAMETER = 72.70
 TOLERANCE LIMIT FACTOR (A Basis) = 8.38
 TOLERANCE LIMIT FACTOR (B Basis) = 4.65
 A-D STATISTIC = 0.97
 OBSERVANCE LEVEL = 0.01

 THE A VALUE = 61.89
 THE B VALUE = 63.41

THE NORMAL METHOD

MEAN = 70.91
 STANDARD DEVIATION = 3.82
 TOLERANCE LIMIT FACTOR (A Basis) = 2.84
 TOLERANCE LIMIT FACTOR (B Basis) = 1.63

A-D STATISTIC	=	1.08
OBSERVANCE LEVEL	=	0.01

THE A VALUE	=	60.05
THE B VALUE	=	64.67

THE LOGNORMAL METHOD

AVERAGE OF LOG DATA	=	4.26
STD. DEV. OF LOG DATA	=	0.05
TOLERANCE LIMIT FACTOR (A Basis)	=	2.84
TOLERANCE LIMIT FACTOR (B Basis)	=	1.63
A-D STATISTIC	=	1.11
OBSERVANCE LEVEL	=	0.01

THE A VALUE	=	60.73
THE B VALUE	=	64.83

THE NONPARAMETRIC METHOD

THE A VALUE	=	57.18	(Hanson-Koopmans value)
THE B VALUE	=	64.90	

NORMAL ANALYSIS IN THE PRESENCE OF BATCH-TO-BATCH VARIATION (ANOVA)

THE ASSUMPTIONS ARE:

- 1) THE DATA FROM EACH BATCH ARE NORMALLY DISTRIBUTED,
- 2) THE BATCH MEANS ARE NORMALLY DISTRIBUTED, AND
- 3) THE VARIANCES WITHIN EACH BATCH ARE THE SAME.

BETWEEN BATCH MEAN SQUARE	=	12.49
WITHIN BATCH VARIANCE	=	14.98
POPULATION STANDARD DEVIATION	=	3.82
TOLERANCE LIMIT FACTOR (A Basis)	=	2.84
TOLERANCE LIMIT FACTOR (B Basis)	=	1.63

THE A VALUE	=	60.07
THE B VALUE	=	64.68

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

B1 Allowee

CASA DENOMINATION **PROBLEM 6 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

No. of batches: **9** ☐ There are different environmental conditions

	Type of data	No. of data	1	2	3	4	5	6	7
1	XXXXXX	7	72.3	65.3	74.3	68.5	65.7	64.3	72.8
2	XXXXXX	5	74.5	66.3	74.3	69.3	64.9		
3	XXXXXX	5	75.3	68.5	75.2	74.5	69.6		
4	XXXXXX	7	65.6	74.6	76.5	72	69.2	74.3	78.7
5	XXXXXX	6	68.1	72.1	66.6	75	67.2	73.4	
6	XXXXXX	5	66.4	72.6	67.5	76.3	65.1		
7	XXXXXX	5	73.3	72.6	67.8	69.4	67.3		
8	XXXXXX	6	75.2	72.8	67.8	76.2	72.8	68.7	
9	XXXXXX	7	72.3	76	65.8	69.9	67.8	74.6	71.3

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.11. Example 6 input data

If the 'Calculate' button is pressed the results are shown:

B1 Allowee

CASA DENOMINATION **PROBLEM 6 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

Allowables: **Statistics of each batch** Results Weibull and Normal Results Lognormal and ANOVA

Batch no.	Sample	Mean	Standar dev.	CoV (%)
1	7	69.03	4.09	5.92
2	5	69.86	4.44	6.35
3	5	72.62	3.30	4.54
4	7	72.99	4.46	6.11
5	6	70.40	3.55	5.04
6	5	69.58	4.71	6.77
7	5	70.08	2.74	3.91
8	6	72.25	3.39	4.69
9	7	71.10	3.61	5.08

No. of batches **9**

Total no. of data **53**

Mean **70.91**

Standar deviation **3.82**

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.12. Example 6 output results

EXAMPLE 7:

The input data file (*ej_allowee_7.dat*) is the following:

DEN CASA: PROBLEM 7 MIL-HDBK-17-1D Section 8.7

DEN MAT.: Z-19776+Z19785

TEST: LAP SHEAR/DIN 65148(RT)

UNITS: MPA

LOTS: 5

CV (%): 2

'XXXXXX', 5, 63.9, 76.2, 76.9, 67.9, 80.4

'XXXXXX', 5, 62.2, 79.8, 62.4, 76.6, 80.5

'XXXXXX', 5, 72.2, 61.3, 74.8, 81.6, 55.9

'XXXXXX', 5, 58.2, 77.0, 87.5, 63.0, 77.4

'XXXXXX', 5, 66.2, 75.8, 62.6, 78.0, 78.4

The output file is:

ej_allowee_7.RES

18-06-2013, Allowee 4.0

CASA DENOMINATION : PROBLEM 7 MIL-HDBK-17-1D Section 8.7
 MATERIAL DENOMINATION : Z-19776+Z19785
 TEST : LAP SHEAR/DIN 65148(RT)
 UNITS : MPA
 TOTAL NUMBER OF VALUES : 25
 NUMBER OF BATCHES : 5

ALLOWABLE VALUES

A-BASIS VALUE = 33.68
 B-BASIS VALUE = 54.67

REMARKS:

Hanson-Koopmans method. The Non parametric method cannot be used, $N < 29$.
 Batches are from the same population. The data may be pooled.
 Batches with similar variances.
 There are not outliers.

INPUT DATA

BATCH	N	SPECIMEN DATA				
1	5	63.9	67.9	76.2	76.9	80.4
2	5	62.2	62.4	76.6	79.8	80.5
3	5	55.9	61.3	72.2	74.8	81.6
4	5	58.2	63.0	77.0	77.4	87.5
5	5	62.6	66.2	75.8	78.0	78.4

TOTAL OF 5 BATCHES, 25 SPECIMEN(S).

PRELIMINARY DESCRIPTIVE STATISTICS

NUMBER OF BATCH	SAMPLE SIZE	AVERAGE	STANDARD DEVIATION
1	5	73.06	6.87
2	5	72.30	9.25
3	5	69.16	10.41
4	5	72.62	11.87
5	5	72.20	7.30

POOLED SAMPLE	25	71.87	8.64
=====			

INVESTIGATION OF BATCH-TO-BATCH VARIABILITY

THE K-SAMPLE ANDERSON-DARLING TEST STATISTIC = 0.59
 THE CRITICAL VALUE = 1.63
 SIGNIFICANT LEVEL = 0.05
 SAME POPULATION

 THE TEST STATISTIC FOR EQUALITY OF VARIANCE = 0.27
 THE CRITICAL VALUE WITH 5-1 DEGREES OF FREEDOM = 2.92
 SIGNIFICANT LEVEL = 0.05
 VARIANCES ARE NOT SIGNIFICANTLY DIFFERENT

=====

COMPUTATION ALLOWABLE VALUES

THE TWO-PARAMETER WEIBULL METHOD

SHAPE PARAMETER = 10.13
 SCALE PARAMETER = 75.57
 TOLERANCE LIMIT FACTOR (A Basis) = 9.50
 TOLERANCE LIMIT FACTOR (B Basis) = 5.20
 A-D STATISTIC = 0.80
 OBSERVANCE LEVEL = 0.03

 THE A VALUE = 50.16
 THE B VALUE = 54.61

THE NORMAL METHOD

MEAN = 71.87
 STANDARD DEVIATION = 8.64
 TOLERANCE LIMIT FACTOR (A Basis) = 3.16
 TOLERANCE LIMIT FACTOR (B Basis) = 1.84
 A-D STATISTIC = 0.95
 OBSERVANCE LEVEL = 0.01

 THE A VALUE = 44.61
 THE B VALUE = 56.00

THE LOGNORMAL METHOD

AVERAGE OF LOG DATA	=	4.27
STD. DEV. OF LOG DATA	=	0.12
TOLERANCE LIMIT FACTOR (A Basis)	=	3.16
TOLERANCE LIMIT FACTOR (B Basis)	=	1.84
A-D STATISTIC	=	1.05
OBSERVANCE LEVEL	=	0.01
THE A VALUE	=	48.35
THE B VALUE	=	56.89

THE NONPARAMETRIC METHOD

THE A VALUE	=	33.68	(Hanson-Koopmans value)
THE B VALUE	=	54.67	(Hanson-Koopmans value)

NORMAL ANALYSIS IN THE PRESENCE OF BATCH-TO-BATCH VARIATION (ANOVA)
THE ASSUMPTIONS ARE:

- 1) THE DATA FROM EACH BATCH ARE NORMALLY DISTRIBUTED,
- 2) THE BATCH MEANS ARE NORMALLY DISTRIBUTED, AND
- 3) THE VARIANCES WITHIN EACH BATCH ARE THE SAME.

BETWEEN BATCH MEAN SQUARE	=	12.02
WITHIN BATCH VARIANCE	=	87.08
POPULATION STANDARD DEVIATION	=	8.49
TOLERANCE LIMIT FACTOR (A Basis)	=	3.16
TOLERANCE LIMIT FACTOR (B Basis)	=	1.84
THE A VALUE	=	45.07
THE B VALUE	=	56.27

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

B1 Allowee

CASA DENOMINATION **PROBLEM 7 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

No. of batches: **5** ☐ There are different environmental conditions

Type of data	No. of data	1	2	3	4	5
1	5	63.9	76.2	76.9	67.9	80.4
2	5	62.2	79.8	62.4	76.6	80.5
3	5	72.2	61.3	74.8	81.6	55.9
4	5	58.2	77	87.5	63	77.4
5	5	66.2	75.8	62.6	78	78.4

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.13. Example 7 input data

If the 'Calculate' button is pressed the results are shown:

B1 Allowee

CASA DENOMINATION **PROBLEM 7 MIL-HDBK-17-1D Section 8.7**

MATERIAL DENOMINATION **Z-19776+Z19785**

TEST **LAP SHEAR/DIN 65148(RT)**

UNITS **MPA**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **2**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

Statistics of each batch | **Results Weibull and Normal** | **Results Lognormal and ANOVA**

A-Basis Value **33.7**

B-Basis Value **54.7**

Remarks

- Hanson-Koopmans method.
- Batches are from the same population.
- Batches with similar variances.
- There are not outliers.

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.14. Example 7 output results

EXAMPLE 8:

The input data file (***ej_allowee_8.dat***) is the following:

```

CASA DENOMINATION      : EXAMPLE 8
MATERIAL DENOMINATION  : Z-19781
TEST                   : TEST
UNITS                  : MPA
LOTS:                  3
CV (%):                2
'XXXXXX',5,118.58,121.77,137.54,140.39,134.03
'XXXXXX',5,133.44,114.56,123.28,130.33,138
'XXXXXX',5,122.69,137.1,137.49,122.87,135.82
    
```

The output file is:

ej_allowee_8.RES

18-06-2013, Allowee 4.0

CASA DENOMINATION : EXAMPLE 8
 MATERIAL DENOMINATION : Z-19781
 TEST : TEST
 UNITS : MPA
 TOTAL NUMBER OF VALUES : 15
 NUMBER OF BATCHES : 3

ALLOWABLE VALUES

B-BASIS VALUE = 112.77

=====

REMARKS:

AGATE method has been used.
 Data fit in a Normal distribution.
 Batches with similar variances.
 There are not outliers.

=====

INPUT DATA

BATCH	N	SPECIMEN DATA				

1	5	118.6	121.8	134.0	137.5	140.4
2	5	114.6	123.3	130.3	133.4	138.0
3	5	122.7	122.9	135.8	137.1	137.5

TOTAL OF 3 BATCHES, 15 SPECIMEN(S).

PRELIMINARY DESCRIPTIVE STATISTICS

NUMBER OF BATCH	SAMPLE SIZE	AVERAGE	STANDARD DEVIATION

1	5	130.46	9.72
2	5	127.92	9.19
3	5	131.19	7.71

POOLED SAMPLE	15	129.86	8.38
---------------	----	--------	------

INVESTIGATION OF BATCH-TO-BATCH VARIABILITY

THE K-SAMPLE ANDERSON-DARLING TEST STATISTIC	=	0.60
THE CRITICAL VALUE	=	1.89
SIGNIFICANT LEVEL	=	0.05
SAME POPULATION		

```

THE TEST STATISTIC FOR EQUALITY OF VARIANCE      =    0.09
THE CRITICAL VALUE WITH  3-1 DEGREES OF FREEDOM  =    5.12
SIGNIFICANT LEVEL                                =    0.05
VARIANCES ARE NOT SIGNIFICANTLY DIFFERENT

```

SMALL SAMPLE ADDITIONAL DATA

Kb (normalized data) = 0.868

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

Type of data	No. of data	1	2	3	4	5
XXXXXX	5	118.58	121.77	137.54	140.39	134.03
XXXXXX	5	133.44	114.56	123.28	130.33	138
XXXXXX	5	122.69	137.1	137.49	122.87	135.82

Figure 7.15. Example 8 input data

If the 'Calculate' button is pressed the results are shown:

B-Basis Value 112.8

Remarks

- AGATE method has been used.
- Data fit in a Normal distribution.
- Batches with similar variances.
- There are not outliers.

Figure 7.16. Example 8 output results

EXAMPLE 9:

The input data file (***ej_allowee_9.dat***) is the following:

DEN CASA: EXAMPLE 9

DEN MAT.: Z-19776

TEST: TEST

UNITS: MPA

LOTS: 2

CV (%): 5.8

'XXXXXX', 2, 237, 234.7

'XXXXXX', 3, 244.4, 242.1, 245.2

3

Y

The output file is:

ej_allowee_9.RES

21-06-2013, Allowee 4.0

CASA DENOMINATION : EXAMPLE 9
 MATERIAL DENOMINATION : Z-19776
 TEST : TEST
 UNITS : MPA
 TOTAL NUMBER OF VALUES : 5
 NUMBER OF BATCHES : 2
 CV(%) OF POPULATION* : 5.8

(*) Coefficient of Variation of population only for small samples formula
 Find recommended values for several data types in user's manual

ALLOWABLE VALUES

ENVIRONMENTAL CONDITION 1
 B-BASIS VALUE = 209.38
 ENVIRONMENTAL CONDITION 2
 B-BASIS VALUE = 216.53

REMARKS:

The data from different environmental conditions are poolable.
 Small samples method has been used.
 Environmental conditions with similar variances.
 The pooled data are assumed to fit in a normal distribution.
 There are not outliers.

INPUT DATA

BATCH	N	SPECIMEN DATA		
1	2	234.7	237.0	
2	3	242.1	244.4	245.2

TOTAL OF 2 BATCHES, 5 SPECIMEN(S).

PRELIMINARY DESCRIPTIVE STATISTICS

NUMBER OF BATCH	SAMPLE SIZE	AVERAGE	STANDARD DEVIATION
1	2	235.85	1.63
2	3	243.90	1.61
POOLED SAMPLE	5	240.68	4.63

INVESTIGATION OF BATCH-TO-BATCH VARIABILITY

THE K-SAMPLE ANDERSON-DARLING TEST STATISTIC = 0.00
 THE CRITICAL VALUE = 2.03
 SIGNIFICANT LEVEL = 0.05
 SAME POPULATION

THE TEST STATISTIC FOR EQUALITY OF VARIANCE = 0.02
 THE CRITICAL VALUE WITH 2-1 DEGREES OF FREEDOM = 758.77
 SIGNIFICANT LEVEL = 0.05
 VARIANCES ARE NOT SIGNIFICANTLY DIFFERENT

SMALL SAMPLE ADDITIONAL DATA

Kb (normalized data) = 0.980
 Coefficient of variation used (%) = 5.8
 Kb (small sample formula) = 0.888

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

Allowee

CASA DENOMINATION **EXAMPLE 9**

MATERIAL DENOMINATION **Z-19776**

TEST **TEST**

UNITS **M**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **5.8**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

No. of batches: **2** ☒ There are different environmental conditions

Type of data	No. of data	1	2	3
1	2	237	234.7	
2	3	244.4	242.1	245.2

Language **Input file** **Output file** **Execute** **Return to data** **Exit**

Figure 7.17. Example 9 input data

If the 'Calculate' button is pressed the results are shown:

Allowee

CASA DENOMINATION **EXAMPLE 9**

MATERIAL DENOMINATION **Z-19776**

TEST **TEST**

UNITS **M**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **5.8**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

Statistics of each batch

Batch no.	Sample	Mean	Standard dev.	CoV (%)
1	2	235.85	1.63	0.69
2	3	243.90	1.61	0.66

No. of batches **2**

Total no. of data **5**

Mean **240.68**

Standard deviation **4.63**

Kb (normalized data) **0.980**

CoV used (%) **5.8**

Kb (SSF) **0.888**

Language **Input file** **Output file** **Execute** **Return to data** **Exit**

Figure 7.18. Example 9 output results

EXAMPLE 10:

The input data file (*ej_allowee_10.dat*) is the following:

DEN CASA: EXAMPLE 10

DEN MAT.: Z-19776

TEST: TEST

UNITS: MPA

BATCHES: 2

CV(%): 9.0

'XXXXXX', 2, 237, 234.7

'XXXXXX', 3, 244.4, 242.1, 245.2

3

Y

The output file is:

ej_allowee_10.RES

21-06-2013, Allowee 4.0

DENOMINACIÓN CASA : EXAMPLE 10
 DENOMINACIÓN DEL MATERIAL : Z-19776
 ENSAYO : TEST
 UNIDADES : MPA
 NÚMERO TOTAL DE VALORES : 5
 NÚMERO DE BATCHES : 2
 CV(%) DE LA POBLACION* : 9.0

(*) Coeficiente de Variacion de población sólo para fórmula de pequeñas muestras
 Valores recomendados para varios tipos de datos en manual de usuario

ADMISIBLES

CONDICIÓN AMBIENTAL 1
 VALOR B = 195.69
 CONDICIÓN AMBIENTAL 2
 VALOR B = 202.37

=====

CONSIDERACIONES:

Los datos de distintas condiciones ambientales se pueden agrupar.
 Utilización de fórmula para pequeñas muestras.
 Igualdad de varianzas entre condiciones ambientales.
 Se asume población normal en los datos agrupados.
 No existen outliers.

=====

DATOS DE ENTRADA

BATCH	N	DATOS DE ESPECÍMENES		
1	2	234.7	237.0	
2	3	242.1	244.4	245.2

TOTAL DE 2 BATCHES, 5 ESPECIMEN(ES).

DATOS ESTADÍSTICOS PRELIMINARES

NÚMERO DE BATCH	TAMAÑO DE MUESTRA	MEDIA	DESVIACIÓN ESTÁNDAR
1	2	235.85	1.63
2	3	243.90	1.61
POOLED SAMPLE	5	240.68	4.63

INVESTIGACION DE LA VARIABILIDAD BATCH-TO-BATCH

VALOR ESTADÍSTICO DEL TEST DE K-SAMPLE ANDERSON-DARLING = 0.00
 VALOR CRÍTICO = 2.03
 NIVEL SIGNIFICATIVO = 0.05
 MISMA POBLACIÓN

VALOR ESTADÍSTICO PARA IGUALDAD DE VARIANZAS = 0.02
 VALOR CRÍTICO CON 2-1 GRADOS DE LIBERTAD = 758.77
 NIVEL SIGNIFICATIVO = 0.05
 VARIANZAS NO SIGNIFICATIVAMENTE DIFERENTES

DATOS ADICIONALES DE PEQUEÑAS MUESTRAS

Kb (datos normalizados) = 0.980
 Coeficiente de variación utilizado (%) = 9.0
 Kb (fórmula pequeñas muestras) = 0.830

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

Allowee

CASA DENOMINATION **EXAMPLE 10**

MATERIAL DENOMINATION **Z-19776**

TEST **TEST**

UNITS **M**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **9**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

No. of batches: **2** ☒ There are different environmental conditions

Type of data	No. of data	1	2	3
1	2	237	234.7	
2	3	244.4	242.1	245.2

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.19. Example 10 input data

If the 'Calculate' button is pressed the results are shown:

Allowee

CASA DENOMINATION **EXAMPLE 10**

MATERIAL DENOMINATION **Z-19776**

TEST **TEST**

UNITS **M**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **9**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

Allowables **Statistics of each batch**

Condition	A Value	B Value
1	----	195.7
2	----	202.4

Remarks

- The data from different environmental conditions are poolable.
- Small samples formula has been used.
- Environmental conditions with similar variances.
- The pooled data are assumed to fit in a normal distribution.
- There are not outliers.

Language

Input file

Output file

Execute **Return to data** **Exit**

Figure 7.20. Example 10 output results

EXAMPLE 11:

The input data file (***ej_allowee_11.dat***) is the following:

DEN CASA: EXAMPLE 11

DEN MAT.: Z-19776

TEST: TEST

UNITS: MPA

BATCHES: 2

CV(%): 0.1

'XXXXXX', 2, 237, 234.7

'XXXXXX', 3, 244.4, 242.1, 245.2

3

Y

The output file is:

ej_allowee_11.RES

21-06-2013, Allowee 4.0

```
DENOMINACIÓN CASA      : EXAMPLE 11
DENOMINACIÓN DEL MATERIAL : Z-19776
ENSAYO                  : TEST
UNIDADES                : MPA
NÚMERO TOTAL DE VALORES : 5
NÚMERO DE BATCHES      : 2
CV(%) DE LA POBLACION*  : 0.1
```

(*) Coeficiente de Variacion de población sólo para fórmula de pequeñas muestras

Valores recomendados para varios tipos de datos en manual de usuario

ADMISIBLES

```
CONDICIÓN AMBIENTAL      1
VALOR B = 233.10
CONDICIÓN AMBIENTAL      2
VALOR B = 241.06
```

CONSIDERACIONES:

Los datos de distintas condiciones ambientales se pueden agrupar.
 Utilización de fórmula para pequeñas muestras.
 Igualdad de varianzas entre condiciones ambientales.
 Se asume población normal en los datos agrupados.
 No existen outliers.

DATOS DE ENTRADA

BATCH	N	DATOS DE ESPECÍMENES		
1	2	234.7	237.0	
2	3	242.1	244.4	245.2

 TOTAL DE 2 BATCHES, 5 ESPECIMEN(ES).

DATOS ESTADÍSTICOS PRELIMINARES

NÚMERO DE BATCH	TAMAÑO DE MUESTRA	MEDIA	DESVIACIÓN ESTÁNDAR
1	2	235.85	1.63
2	3	243.90	1.61
POOLED SAMPLE	5	240.68	4.63

INVESTIGACION DE LA VARIABILIDAD BATCH-TO-BATCH

VALOR ESTADÍSTICO DEL TEST DE K-SAMPLE ANDERSON-DARLING = 0.00
 VALOR CRÍTICO = 2.03
 NIVEL SIGNIFICATIVO = 0.05
 MISMA POBLACIÓN

 VALOR ESTADÍSTICO PARA IGUALDAD DE VARIANZAS = 0.02
 VALOR CRÍTICO CON 2-1 GRADOS DE LIBERTAD = 758.77
 NIVEL SIGNIFICATIVO = 0.05
 VARIANZAS NO SIGNIFICATIVAMENTE DIFERENTES

DATOS ADICIONALES DE PEQUEÑAS MUESTRAS

Kb (datos normalizados) = 0.980
 Coeficiente de variación utilizado (%) = 0.6
 Kb (fórmula pequeñas muestras) = 0.988

If the PC version is used and the input file shown before is open, the program shows the following on the screen:

Allowee

CASA DENOMINATION **EXAMPLE 11**

MATERIAL DENOMINATION **Z-19776**

TEST **TEST**

UNITS **M**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **0.1**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

No. of batches: **2** ☒ There are different environmental conditions

Type of data	No. of data	1	2	3
1	2	237	234.7	
2	3	244.4	242.1	245.2

Language **Input file** **Output file** **Execute** **Return to data** **Exit**

Figure 7.21. Example 11 input data

If the 'Calculate' button is pressed the results are shown:

Allowee

CASA DENOMINATION **EXAMPLE 11**

MATERIAL DENOMINATION **Z-19776**

TEST **TEST**

UNITS **M**

COEFFICIENT OF VARIATION OF POPULATION, CV(%) **0.1**

Values to calculate

☐ Only from laboratory

☐ Only from qualification

☒ All

Allowables **Statistics of each batch**

Condition	A Value	B Value
1	----	233.1
2	----	241.1

Remarks

- The data from different environmental conditions are poolable.
- Small samples formula has been used.
- Environmental conditions with similar variances.
- The pooled data are assumed to fit in a normal distribution.
- There are not outliers.

Language **Input file** **Output file** **Execute** **Return to data** **Exit**

Figure 7.22. Example 11 output results